# MATH 301: Homework 5

Aren Vista

March 26, 2026

Section 3.4 : Use method of *Example 3.4.3(b)* to show $0 < c < 1 \implies \lim c^{\frac{1}{n}} = 1$

Example (b): Proving $\lim(c^{1/n}) = 1$ for $c > 1$

**The Goal:** Prove that the $n$-th root of a number greater than 1 gets closer and closer to 1 as $n$ gets larger. Let our sequence be $z_n = c^{1/n}$.

**Step 1: Show the sequence is decreasing**
The text asks "(Why?)" here. Here is the reason: Since $c > 1$, taking a smaller and smaller fractional power of $c$ yields a smaller and smaller result. Because $\frac{1}{n+1} < \frac{1}{n}$, it follows that:

$$c^{1/(n+1)} < c^{1/n}$$

$$z_{n+1} < z_n$$

So, the sequence is decreasing.
**Step 2: Show the sequence is bounded**
Because $c > 1$, any positive root of $c$ will still be greater than 1.

$$z_n > 1$$

The sequence is bounded below by 1.
**Step 3: Conclude it converges**
Just like in example (a), we have a decreasing sequence that is bounded below. By the Monotone Convergence Theorem, it converges. Let's call the limit $z$:

$$z = \lim(z_n)$$

**Step 4: Find the limit using the same trick**
Let's look at the subsequence of even terms, $z_{2n}$. Because the main sequence converges to $z$, this subsequence also converges to $z$. Let's look at the algebra:

$$z_{2n} = c^{1/(2n)} = (c^{1/n})^{1/2} = \sqrt{z_n}$$

If we square both sides to get rid of the fraction, we get:

$$(z_{2n})^2 = z_n$$

Now, take the limit of both sides as $n \to \infty$:

$$z^2 = z$$

**Step 5: Solve for $z$**
Again, the only solutions to $z^2 = z$ are $z = 0$ and $z = 1$. Since we established in Step 2 that every term in our sequence is strictly greater than 1 ($z_n > 1$), the limit cannot possibly be 0. Therefore, the limit must be 1.

$$\lim(c^{1/n}) = 1$$

Section 3.4 : Use method of *Example 3.4.3(b)* to show $0 < c < 1 \implies \lim c^{\frac{1}{n}} = 1$

$$0 < c < 1 \implies \exists\ a < b, \in \mathbb{R} \text{ s.t. } c := \frac{a}{b}$$

$$\text{Observe: } 0 < c < 1 \implies \lim_{n \to \infty} c^n = 0 \implies \forall\ n \in \mathbb{N}, c_n \in (0,1] \ \wedge \ c^n < c^{n+1}$$

$$c^n \cdot c^{\frac{1}{n^2}} < c^{n+1} \cdot c^{\frac{1}{n^2}}$$

$$c^{\frac{1}{n}} < c^{\frac{1}{n} + \frac{1}{n^2}}$$

$$\therefore c^{\frac{1}{n}} < c^{\frac{1}{n+1}}$$

$$\therefore\ c^{n+1} < c^n < c^{\frac{1}{n}} < c^{\frac{1}{n+1}} \implies c^n \text{ is monotonic and increasing.}$$

Let $c_{2n}$ be a subsequence of $c_n$ containing even naturals

$$z_{2n} := \left(\frac{a}{b}\right)^{\frac{1}{2n}} \quad z_n := \left(\frac{a}{b}\right)^{\frac{1}{n}}$$